Aaryahi Ahluwalia

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B.Tech CS-AI student building **explainable ML for financial risk**: fraud and scam detection, credit scoring, LLM verification.

Education

Banasthali Vidyapith Jul 2024 – May 2028
B.Tech, Computer Science & Artificial Intelligence *CGPA: 7.6/10* *Niwai, Rajasthan*

Experience

Origen Technologies Jun 2026 – Jul 2026
Software Engineer Intern *Abu Dhabi, UAE*

- Authored the **Integration Design Document** for a UAE government Smart Parks platform: a 6-layer digital-twin architecture linking **1,400+ LoRaWAN/4G sensors** across 6 parks (parking, waste, CO_2, CCTV, energy, water).
- Specified **9 REST API integration sequences with JWT auth** (device registration, GeoJSON spatial ingestion, indoor mapping, alert triggering, live dashboards) across a multi-vendor inventory incl. Milesight and Dahua AI CCTV.
- Bridged Chinese IoT vendors and UAE stakeholders; spec shipped with a risk register, rated "well beyond" intern level.

ISIT Global May 2025 – Jul 2025
AI/Cloud Implementation Intern *Dubai, UAE*

- Designed Azure generative-AI use cases (developer copilots, knowledge assistants, document intelligence, AIOps) for Etimad Holding's enterprise transformation programme.
- Produced architecture artifacts and proof-of-concept plans for a UAE Government hazard-identification system, aligned with national safety and compliance frameworks.

Projects

RiskPulse: Real-Time Transaction Risk & Scam-Coercion Engine GitHub
Python, FastAPI, scikit-learn, SHAP, NetworkX, Redis, PostgreSQL, Docker, React

- Scores each transaction on **30+ engineered features** (IEEE-CIS data) with a calibrated supervised + Isolation Forest ensemble, **per-decision SHAP explanations** and 3-tier adaptive verification (approve / step-up / block).
- Built **"puppet" coercion detection for digital-arrest scams**: 4 behavioural signals (amount and timing regularity, new-payee bursts, session linearity) catch victims scripted through payments they authorise themselves, where login and 2FA checks all pass.
- Ops-ready: JWT/OAuth2 API, Redis feature store with p95/p99 latency tracking, NetworkX graph fraud-contagion scoring, champion/challenger retraining with rollback; tests caught and fixed a silent Redis-failover bug.

ArthSetu: Explainable Credit Scoring for Thin-File Borrowers Live
Python, scikit-learn, SHAP, React, Supabase (PostgreSQL)

- Logistic-regression credit-risk model on alternative data (**AUC 0.77**) with **SHAP reason codes**, so each applicant sees why they scored as they did and what would improve it.
- Applicant and lender dashboards with role-based access enforced by PostgreSQL Row-Level Security.

RedShift: Forensic Red-Flag & Market-Mismatch Detector
Python, LLMs, RAG, pytest

- Mines Indian annual reports for forensic red flags (receivables vs revenue, auditor exits, promoter pledging, related-party deals) where the stock still trades calm; back-testing on Yes Bank, DHFL and IL&FS.
- Built the **LLM validation layer**: checks every figure and quote the LLM cites against the source filing, verbatim, before it reaches a user.

Agastya: Physics-Informed Exoplanet Detection GitHub
Python, PyTorch, SHAP, FastAPI, React

- 5-stage pipeline on Kepler/TESS light curves: denoising, TLS/BLS transit search, 4-check astrophysical vetting, TCN-Transformer classifier, SHAP + attention explanations; human-in-the-loop review UI.

+ More projects on my portfolio.

Technical Skills

Languages: Python, Java, SQL, C, MATLAB
ML & Explainability: scikit-learn, XGBoost, Random Forest, PyTorch, SHAP, Isolation Forest, SMOTE, calibration, TCN-Transformer, YOLO, MediaPipe, computer vision, Pandas, NumPy
LLMs & Agents: RAG, LLM output verification, agentic workflows, Llama 3, Groq, Ollama
Backend & Data: FastAPI, Flask, REST, JWT/OAuth2, PostgreSQL, Redis, SQLite, Supabase, NetworkX, Streamlit, Plotly
Cloud & Systems: Docker, Git, Linux (shell scripting, system calls), Azure (**AZ-900**), pytest, ROS 2, IoT (LoRaWAN)
Architecture: solution and pipeline diagrams in Mermaid and PlantUML, API integration design documents

Achievements & Leadership

- **Barclays Hackathon Finalist** (ArthSetu) · **2nd Place, Udgam Hackathon**
- **Top 10 National Finalist, Finvesta '26** (St. Xavier's) · **Amazon ML Challenge 2026:** 0.967
- **Software Lead, NIDAR AirMouse 2026–27:** leading 10 engineers on a GPS-denied autonomous rescue drone.
- **Former national-level basketball player** · English, Hindi, Punjabi, Arabic · **UAE Golden Visa holder**